

Side Story I: Chern’s Intrinsic Proof of Gauss–Bonnet Theorem on Closed Orientable Manifolds

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Preface

This note is written as a side story of the informal summer 2026 seminar on vector bundles and characteristic classes. The seminar aims to provide a self-contained introduction to the theory of vector bundles in various aspects.

This note is to give an expository introduction to the famous work [1] by Shing-Shen Chern, which gives an intrinsic proof of the Gauss–Bonnet theorem, also known as the Chern–Gauss–Bonnet theorem today, on closed orientable manifolds.

In the entire note, for simplicity of notations, we use the summation convention that repeated indices represent summation.

1 Preliminary on Differential Geometry

1.1 Riemannian Manifolds

1.2 Connection and Curvature

Definition 1.1 (Connection).

The notion of connection can be canonically extended to all degrees of TM -valued differential forms, called the covariant derivative

Definition 1.2 (Covariant Derivative).

Definition 1.3 (Curvature).

A special case of the connection on a Riemannian manifold (M, g) is the Levi–Civita connection:

Definition 1.4 (Levi–Civita Connection and Riemannian Curvature).

Theorem 1.5 (Bianchi Identities).

1.3 Principal Bundles and Moving Frames

Theorem 1.6 (Structure Equations). *Consider the orthonormal frame $e_1, \dots, e_n$ and the corresponding coframe $\theta_1, \dots, \theta_n$. The following identity holds*

$$(SE) \quad \begin{cases} T_i^\nabla = d\theta_i + \omega_{ij} \wedge \theta_j, \\ F_{ij}^\nabla = d\omega_{ij} + \omega_{ik} \wedge \omega_{kj}, \end{cases}$$

where ∇ is the chosen connection, $\omega \in \Omega^1(M, \mathfrak{g})$ be the connection 1-form, $T^\nabla \in \Omega^2(M)$ be the torsion tensor, and $F^\nabla \in \Omega^2(M, \mathfrak{g})$ be the curvature 2-form.

Proposition 1.7 (Change of Frame). *Under the change of frame $\tilde{e}_j = e_i a_{ij}$, the change of connection 1-form and curvature are given by*

$$\tilde{\omega}_{ij} = a_{ik}^{-1} \omega_{kl} a_{lj} + a_{ik}^{-1} da_{kj}, \quad \tilde{\Omega}_{ij}^\nabla = a_{ik}^{-1} \Omega_{kl}^\nabla a_{lj}.$$

1.4 De Rham Cohomology

1.5 Poincaré–Hopf Theorem

2 Proof of the Theorem

We can now dive into Chern’s proof of the theorem. Since the formulation of differential geometry has changed greatly in the 20th century, we will not follow the mathematical language in the original paper [1]; instead, we may use modern notations and aspects to reformulate the proof.

2.1 Setting and Statement of the Theorem

Let (M, g) be an n -dimensional closed oriented Riemannian manifold with Riemannian metric g and $n = 2m$ for some m . In the following passages, all the connections we use are Levi–Civita connection $\nabla := \nabla^{TM}$ with curvature 2-form, denoted by $\Omega^\nabla \in \Omega^2(M, \mathfrak{so}(n))$. Under this setting, the goal of the paper is to intrinsically prove the following theorem:

Theorem 2.1 (Chern–Gauss–Bonnet Theorem). *The Euler characteristic of M satisfies the following identity*

$$(CGB) \quad \chi(M) = \int_M e(\nabla^{TM}) = \frac{1}{(2\pi)^m} \int_M \text{Pf}(\Omega^\nabla),$$

where $e(\nabla^{TM}) := \text{Pf}(\Omega^\nabla/2\pi)$ is called the Euler form.

The aim of Chern's paper is not to reprove a specific embedding theorem but to provide a direct, intrinsic proofone that avoids embedding the Riemannian manifold into Euclidean space and instead relies directly on Riemannian curvature forms and vector fields. The introduction of Chern's paper notes that related work by Allendoerfer, Fenchel, and AllendoerferWeil still relied on Euclidean embeddings or the embedding of Riemannian cells; Cherns objective is to move beyond this approach.

2.2 Outline of the Proof

The proof by Chern can be summarized into the following steps:

Firstly, define the Euler form on the base manifold M using the Pfaffian as an $\text{SO}(n)$ -invariant polynomial $\text{Pf} \in I^m(\text{SO}(n))$:

$$(Eu) \quad e(\nabla^{TM}) := \text{Pf} \left(\frac{\Omega^\nabla}{2\pi} \right) = \frac{1}{2^m m! (2\pi)^m} \epsilon_{i_1 \dots i_{2m}} \Omega_{i_1 i_2}^\nabla \wedge \dots \wedge \Omega_{i_{2m-1} i_{2m}}^\nabla,$$

where $\Omega^\nabla := (\Omega_{ij}^\nabla)_{1 \leq i, j \leq 2m}$ is the curvature 2-form. We need to show that this form is invariant under the change of local frame, i.e., invariant under the $\text{SO}(n)$ -action.

Secondly, consider the unit sphere bundle $\pi : S(TM) \rightarrow M$, where $\dim S(TM) = 4m - 1$ defined by

$$S(TM) := \{(x, v) \in TM \mid \|v\| = 1\}.$$

Since the unit sphere bundle is locally trivial, for $v \in S_x M$, one can get a tautological vector field $u_{(x,v)} = v \in T_x M$ as the section of the pullback bundle $\pi^* TM \rightarrow S(TM)$. With $v := u_i e_i$ for frame $e_i \in T_x M$, we may define the covariant derivative of the tautological vector field

$$\sum_i \eta_i \otimes \pi^* e_i = \nabla^{\pi^* TM} U$$

such that $\sum_i u_i \eta_i = 0$, and normalized on fibers.

Thirdly, construct the transgression form $\mathcal{T}_\nabla$ recursively using the information we have in the second step, such that $d\mathcal{T}_\nabla = \pi^* e(\nabla^{TM})$.

Finally, use Stokes' formula and Poincaré–Hopf theorem to connect the integration of the Euler form to the Euler characteristic, which completes the proof.

The working space in the proof can be summarized in the following diagram:

$$\begin{array}{ccc} \pi^* TM & \xrightarrow{\text{pr}_2} & TM \\ u \uparrow \downarrow \text{pr}_1 & \nearrow \iota & \uparrow \downarrow \pi \\ S(TM) & \xleftarrow[\text{s}_X]{\pi|_{S(TM)}} & M \end{array}$$

where $u \in \Gamma(S(TM), \pi^* TM)$ is the tautological section, and $s_X(x) := (x, X/\|X\|)$ for $x \in M \setminus Z(X)$ is the unite bundle corresponding to the vector field $X \in \Gamma(TM)$ with isolated zeros on M , where $Z(X) := \{x \in M \mid X(x) = 0\}$ is the set of zeros of X .

2.3 Detail of Shing-Shen Chern's Proof

Let us start with basic linear algebra.

Definition 2.2 (Pfaffian). Let $A = (A_{ij})_{1 \leq i, j \leq 2m}$ be a skew-symmetric matrix. Its Pfaffian is defined by

$$(Pf) \quad Pf(A) := \frac{1}{2^m m!} \epsilon_{i_1, \dots, i_{2m}} A_{i_1 i_2} A_{i_3 i_4} \cdots A_{i_{2m-1} i_{2m}}.$$

The most direct observation of this definition is that

1. $Pf(A)^2 = \det(A)$ is a degree m homogeneous polynomial.
2. For any $a \in O(2m)$, the Pfaffian satisfies $Pf(a^{-1} A a) = \det(a) Pf(A)$. This implies that the Pfaffian is $SO(2m)$ -invariant, i.e., $Pf(a^{-1} A a) = Pf(A)$ for any $a \in SO(2m)$.

Since the tangent bundle $TM \rightarrow M$ on the orientable manifold M has the structure group $SO(n)$ with $n = 2m$, we can define an invariant polynomial on curvature form as Eq. (Eu):

$$(Eu) \quad e(TM) := Pf \left(\frac{\Omega^\nabla}{2\pi} \right) = \frac{1}{2^m m! (2\pi)^m} \epsilon_{i_1 \dots i_{2m}} \Omega_{i_1 i_2}^\nabla \wedge \cdots \wedge \Omega_{i_{2m-1} i_{2m}}^\nabla,$$

Under the change of frames $e'_j = e_i a_{ij}$ for $a \in SO(2m)$, one can check with the change of frame formula in Proposition 1.7 that $Pf(\Omega)$ is indeed invariant. Another observation is that under our setting of the manifold M , the Euler form is closed but not exact, since it is the scalar multiplier of the volume form.

Also, recall that for Levi-Civita connection $\nabla := \nabla^{TM}$ we defined in Definition 1.4 on TM is the connection that is compatible with the Riemannian metric and torsion-free. Let $\omega_{ij} \in \Omega^1(M)$ be the connection 1-form such that for an orthonormal frame $e_1, \dots, e_n$ and corresponding coframe $\theta_1, \dots, \theta_n$ on an open set U ,

$$\nabla e_j = e_i \otimes \omega_{ij} \in \Omega^1(M, TM) \cong \Gamma(TM) \otimes_{C^\infty(M)} \Omega^1(M).$$

Cartan's method of moving frames gives the structure equations for Levi-Civita connection as a special case of Eq.(SE) in Theorem 1.6

$$(SE') \quad \begin{cases} d\theta_i + \omega_{ij} \wedge \theta_j = 0, \\ \Omega_{ij}^\nabla = d\omega_{ij} + \omega_{ik} \wedge \omega_{kj}, \end{cases}$$

where the first structure equation is just the torsion-free condition, and the second structure equation is directly from the definition of curvature. As we have introduced in Definition 1.2, the connection naturally extends to any degree of differential forms $D := d + \omega : \Omega^k(M, TM) \rightarrow \Omega^{k+1}(M, TM)$ defined by $D(\alpha \otimes s) := d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s$.

Remark. Note that the first structure equation in Eq. (SE') is just $D\theta = 0$, the second structure equation can be rewritten as $\Omega^\nabla = D^2$, and the Bianchi identity in Theorem 1.5 gives $D\theta = \Omega^\nabla \wedge \theta$ and $D\Omega^\nabla = 0$.

Defined the unit sphere bundle $S(TM) \xrightarrow{\pi} M$ be given by the total space

$$S(TM) := \{v_x \in TM \mid \|v_x\| = 1, x \in X\}$$

with the projection defined by the restriction of the projection on TM . We have mentioned that the dimension of the total space is given by

$$\dim S(TM) = 2 \dim(M) - 1 = 4m - 1.$$

Note that the unit sphere bundle can be defined as the associated bundle of the oriented orthonormal frame bundle: $F_{SO}M \times_{SO(2m)} S^{2m-1} \rightarrow M$, where $SO(2m) \curvearrowright S^{2m-1}$ as its standard representation. This aspect provided us with a modern understanding of the Euler form (or the Euler class, for the corresponding de Rham cohomology class) using the Chern-Weil theory on the principal $SO(2m)$ -bundle $F_{SO}M \rightarrow M$.

Structure on unit sphere bundle. We consider the pullback vector bundle $\pi^*TM \rightarrow S(TM)$ given by

$$\pi^*TM := \{((x, v), w) \in S(TM) \times TM \mid \pi(v) = \pi(w)\},$$

which the fiber of π^*TM at (x, v) is just $(\pi^*TM)_{(x, v)} = T_xM$. Note that π^*TM admits a pullback connection $\nabla = \pi^*\nabla^{TM}$. In the following passage of the proof, for the simplicity of the notation, we may slightly abuse the notation and not distinguish the notation of objects on TM and on the pullback bundle π^*TM . On this pullback bundle, we can find a tautological unit section $u \in \Gamma(S(TM), \pi^*TM)$ such that $u_{(x, v)} = v$. The tautological bundle, of course, satisfies the unit length condition

$$\langle u, u \rangle_{(x, v)} \equiv 1 \quad \forall (x, v) \in S(TM).$$

We may also write this formula in a choice of orthonormal frame $e_1, \dots, e_n$ in T_xM . Let $u = u_i e_i$, then the condition is just $u_i u_i = (u_1)^2 + \dots + (u_n)^2 = 1$.

Remark. From a modern perspective, for some vector bundle $E \rightarrow X$ which admits a bundle metric and a connection ∇^E , note that due to the definition of $S(E)$, every section $s : X \rightarrow S(E)$ must be non-vanishing and leads to a non-vanishing section $\tilde{s} : X \rightarrow E$ by composite with the inclusion map $\iota : S(E) \hookrightarrow E$.

Thus, with the tautological section u , the pullback bundle on the unit sphere bundle $\pi^*E \rightarrow S(E)$ has a canonical splitting

$$\pi^*E = \langle u \rangle \oplus u^\perp$$

where $\langle u \rangle$ is the trivial line bundle generated by u , the triviality is obtained by the fact that $u_{(x, v)} \neq 0$ for any (x, v) . Then, there are two naturally existing connections: The connection that preserves the splitting $\nabla^0 := d_{\langle u \rangle} \oplus \nabla^{u^\perp}$ and the pullback connection $\nabla^1 := \pi^*\nabla^E$. Note that consider section $s = fu + \xi$

$$\nabla^0(fu + \xi) = df \otimes u + \text{pr}_{u^\perp}(\nabla^0 \xi),$$

and $\nabla^0 u = d1 \otimes u = 0$. Thus, the curvature matrix has an entire column corresponding to $\langle u \rangle$ being zero. Thus, $e(\nabla^0) = \text{Pf}(\Omega^{\nabla^0}/2\pi) = 0$. By Chern-Weil theorem, we will expect the existence of a transgression for given by some Chern-Simons form such that

$$\pi^*e(\nabla^E) = e(\nabla^1) - e(\nabla^0) = d\text{CS}_{\text{Pf}}(\nabla^0, \nabla^1),$$

and the transgression $\mathcal{T}_\nabla$ we want is exactly this Chern-Simons form.

As we mentioned before, we can still denote the pullback connection and the pullback covariant derivative as ∇ and D , where for $\alpha \otimes s \in \Omega^k(S(TM), \pi^*TM)$

$$D(\alpha) = d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s.$$

Using the covariant derivative, we defined the 1-form of the infinitesimal change of the unit vector relative to parallel transportation

$$\eta := Du = du_i \otimes e_i + \omega_{ij} u_j \otimes e_i \in \Omega^1(S(TM), \pi^*TM).$$

We may illustrate the geometrical meaning of this form by considering the smooth curve $\gamma(t) = (x(t), v(t)) \in S(TM)$ for $v(t) = u^i(t)e_i(t) \in T_{x(t)}M$ and $\|v(t)\| \equiv 1$. We may treat $\gamma : I \rightarrow S(TM)$ as a vector field along $x(t)$, then

$$\begin{aligned} \eta(\dot{\gamma}(t)) &= du_i(\dot{\gamma}(t))e_i(t) + \omega_{ij}(\dot{x}(t))u_j(t)e_i(t) \\ &= \left(\frac{du_i}{dt} + \omega_{ij}(\dot{x}(t))u_j(t) \right) e_i(t) = \nabla_{\dot{\gamma}(t)} v(t) \end{aligned}$$

Thus, η is indeed measuring the infinitesimal change of u under parallel transportation. There are two important cases of this fact that we need to emphasize:

1. Suppose $\gamma(t) \in S_x M$ stays in the single fiber for any time t . Then $x(t) \equiv x$ for any t , and this implies $\dot{x} = 0$, i.e.,

$$\eta_i(\dot{\gamma}) = du_i(\dot{\gamma}) + \omega_{ij}(\dot{x})u_i = du_i(\dot{\gamma}),$$

and thus, for any $x \in M$, the restriction to fiber $\eta_i|_{S_x M} = du_i$.

2. Suppose $v(t)$ parallel transported along $x(t)$, then $\eta(\dot{\gamma}) = \nabla_{\dot{x}} v = 0$.

We need to remark on the property of this vector-valued 1-form η that

Proposition 2.3. *The covariant derivative $\eta = Du$ is orthogonal to u , i.e., $\langle u, Du \rangle = 0$.*

Proof. We can prove directly by computation: Note that D is the pullback of Levi-Civita connection, which is defined to be compatible with the pullback metric

$$d\langle u, u \rangle = \langle Du, u \rangle + \langle u, Du \rangle = 2\langle u, Du \rangle$$

Since u is a unit vector field, $\langle u, u \rangle \equiv 1$ and thus the equation above is 0. □

It is also clear in local coordinates: With the chosen orthonormal frame,

$$\begin{aligned} \langle u, Du \rangle &= u_i \eta_i = u_i du_i + u_i \omega_{ij} u_j \\ &= \frac{1}{2} d(u_i u_i) + u_i \omega_{ij} u_j. \end{aligned}$$

The first term is 0 since $u_i u_i \equiv 1$ is constant, and since $\omega_{ij} + \omega_{ji} = 0$, the second term is also zero. This proposition implies that η only measures the information on the orthogonal bundle $u^\perp$ of the tautological bundle $\langle u \rangle$ generated by the tautological section u , i.e., $\eta \in \Omega^1(S(TM), u^\perp)$, where

$$u_{(x,v)}^\perp := \{w \in T_x M \mid \langle w, v \rangle = 0\},$$

and there is a natural isomorphism $T_v S_x M \cong u_{(x,v)}^\perp$. By taking the covariant derivative again on η , we obtain the following proposition:

Proposition 2.4. *The covariant derivative of η satisfies $D\eta = \Omega^\nabla u$.*

Note that $\Omega^\nabla \in \Omega^2(S(TM), \text{End}(\pi^* TM))$ is $\text{End}(\pi^* TM)$ -valued 2-form, which can safely act on $u \in \Gamma(S(TM), \pi^* TM)$. We now prove the proposition:

Proof of Proposition 2.4. The proof is also by the straightforward computation:

$$\begin{aligned} D\eta &= D(\eta_i \otimes e_i) = d\eta_i \otimes e_i + (\eta_i \wedge \omega_{ij}) \otimes e_j \\ &= d\omega_{ij} u_j \otimes e_i - (\omega_{ij} \wedge du_j) \otimes e_i + (du_i \wedge \omega_{ij}) \otimes e_j + (\omega_{ik} \wedge \omega_{kl} u_l) \otimes e_i \\ &= (d\omega_{ij} + \omega_{ik} \wedge \omega_{kj}) u_j \otimes e_i = \Omega_{ij}^\nabla u_j \otimes e_i \end{aligned}$$

This completes the proof of the proposition. □

We also need to check the transformation law of u and η under a change of frame. Let $\tilde{e}_j = e_i a_{ij}$, for $a : U \rightarrow \text{SO}(2m)$, then

$$u_i e_i = \tilde{u}_i \tilde{e}_i = \tilde{u}_i a_{ji} e_i = \tilde{u}_i a_{ij}^{-1} e_i.$$

Thus, $\tilde{u}_i = a_{ij}^{-1} u_j$, note that in Proposition 1.7, we have shown that the connection 1-form transform as

$$\tilde{\omega}_{ij} = a_{ik}^{-1} \omega_{kl} a_{lj} + a_{ik}^{-1} da_{kj}.$$

We may use these formulas to find the transformation law of η :

Proposition 2.5. *The change of frame formula for η is given by $\tilde{\eta}_i = a_{ij}^{-1}\eta_j$*

Proof. The proof is just computations:

$$\begin{aligned}
 (*) \quad \tilde{\eta}_i &= d\tilde{u}_i + \tilde{\omega}_{ij}\tilde{u}_j \\
 &= (da_{ij}^{-1}u_j + a_{ij}^{-1}du_j) + (a_{ik}^{-1}\omega_{kl}a_{lj} + a_{ik}^{-1}da_{kj})a_{jr}^{-1}u_r
 \end{aligned}$$

Since any $a \in \text{SO}(2m)$ satisfies $a^T a = I$, by take the exterior derivative $d(a_{ij}^{-1}a_{jk}) = da_{ij}^{-1}a_{jk} + a_{ij}^{-1}da_{jk} = d\delta_{ik} = 0$, we obtain $da_{ij}^{-1} = -a_{ik}^{-1}da_{kl}a_{lj}^{-1}$, and the Eq.(*) can be rewrite as

$$\begin{aligned}
 \tilde{\eta} &= -a_{ik}^{-1}da_{kl}a_{lj}^{-1}u_j + a_{ij}^{-1}du_j + (a_{ik}^{-1}\omega_{kl}a_{lj} + a_{ik}^{-1}da_{kj})a_{jr}^{-1}u_r \\
 &= (a_{ik}^{-1}da_{kj}a_{jr}^{-1} - a_{ik}^{-1}da_{kj}a_{jr}^{-1})u_r + a_{ik}^{-1}(du_k + \omega_{kl}a_{lj}a_{jr}^{-1}u_r) \\
 &= a_{ij}^{-1}(du_j + \omega_{jk}u_k) = a_{ij}^{-1}\eta_j.
 \end{aligned}$$

This shows the transformation law that $\tilde{\eta}_i = a_{ij}^{-1}\eta_j$. □

References

- [1] Shiing-shen Chern. “On the Curvatura Integra in a Riemannian Manifold”. In: *Annals of Mathematics* 46.4 (1945), pp. 674–684. ISSN: 0003486X, 19398980. URL: <http://www.jstor.org/stable/1969203> (visited on 07/03/2026).